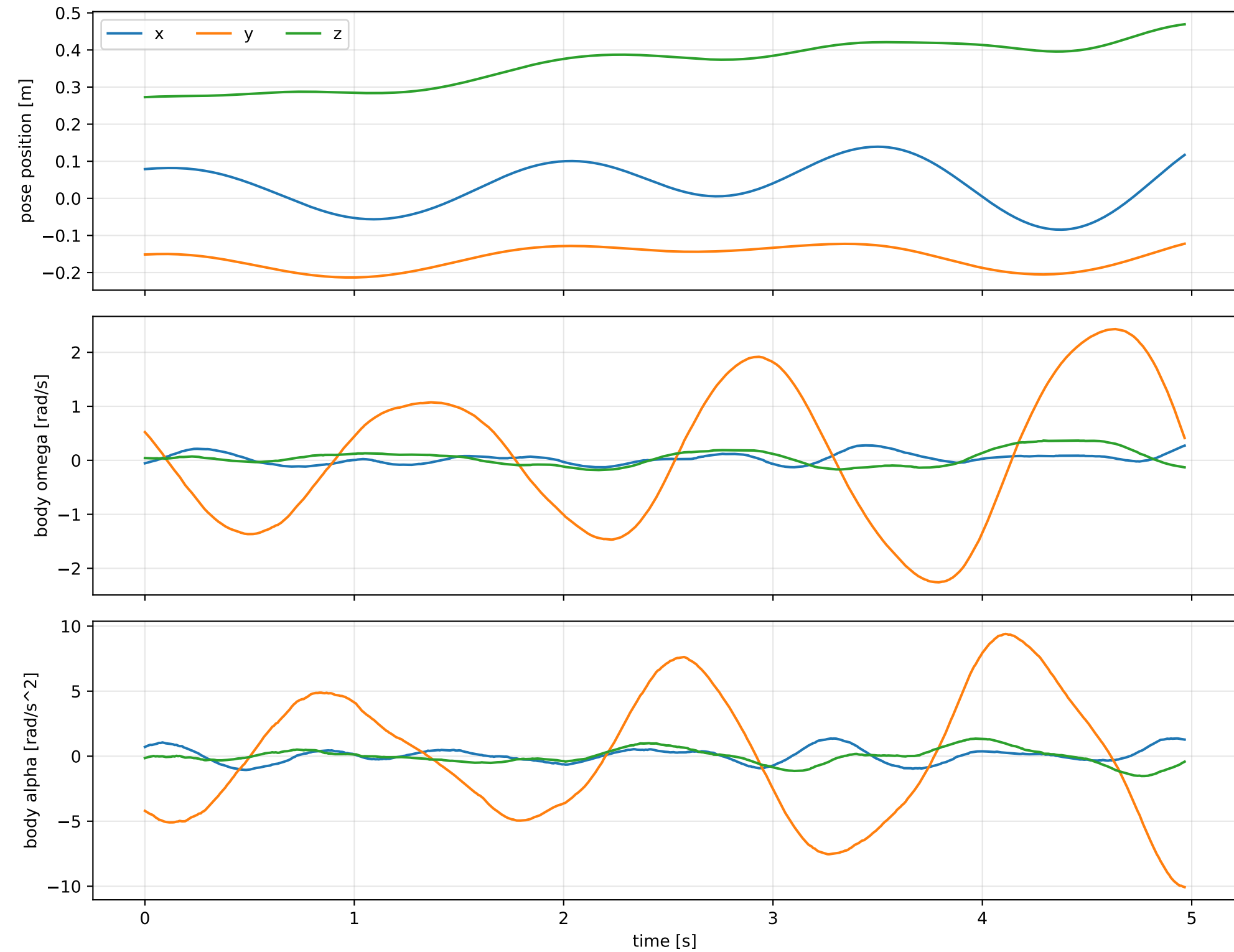
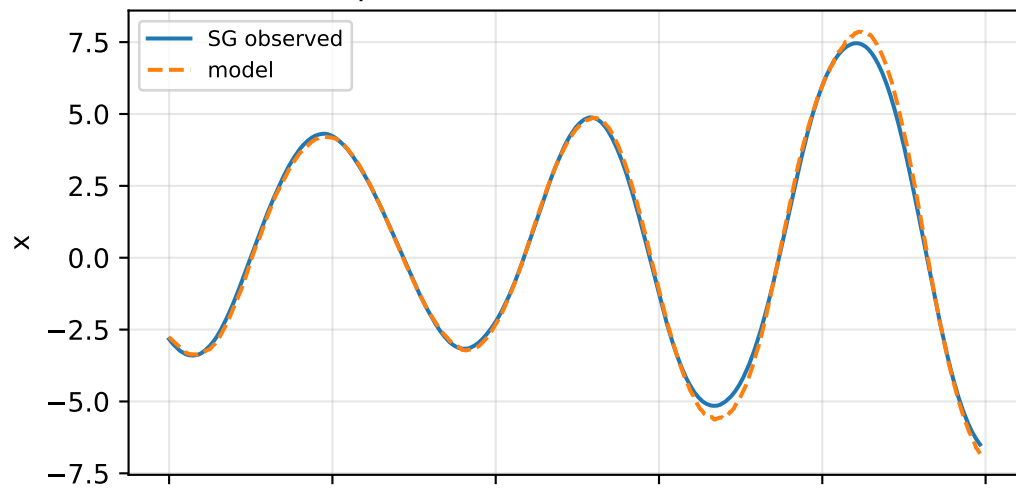
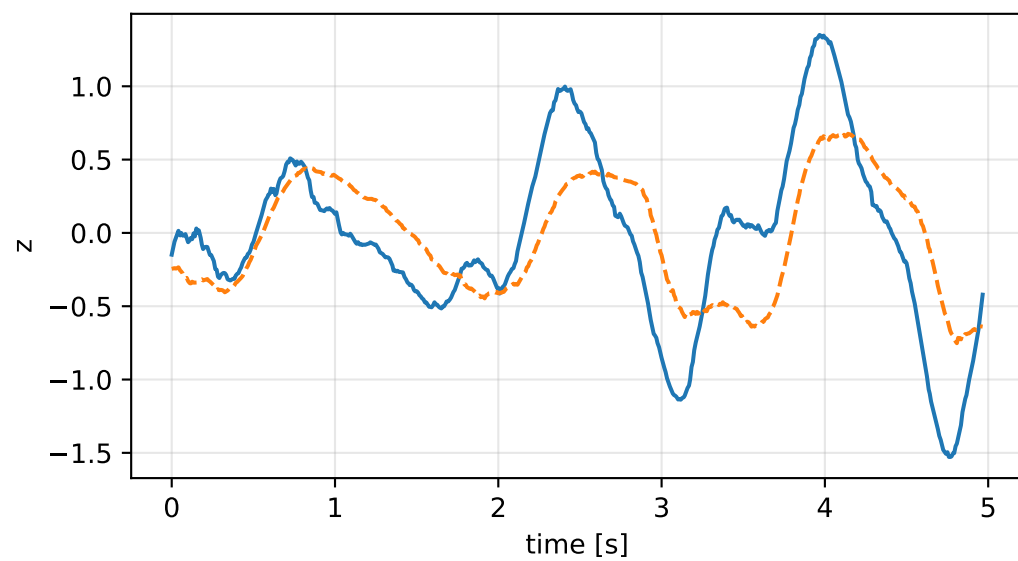
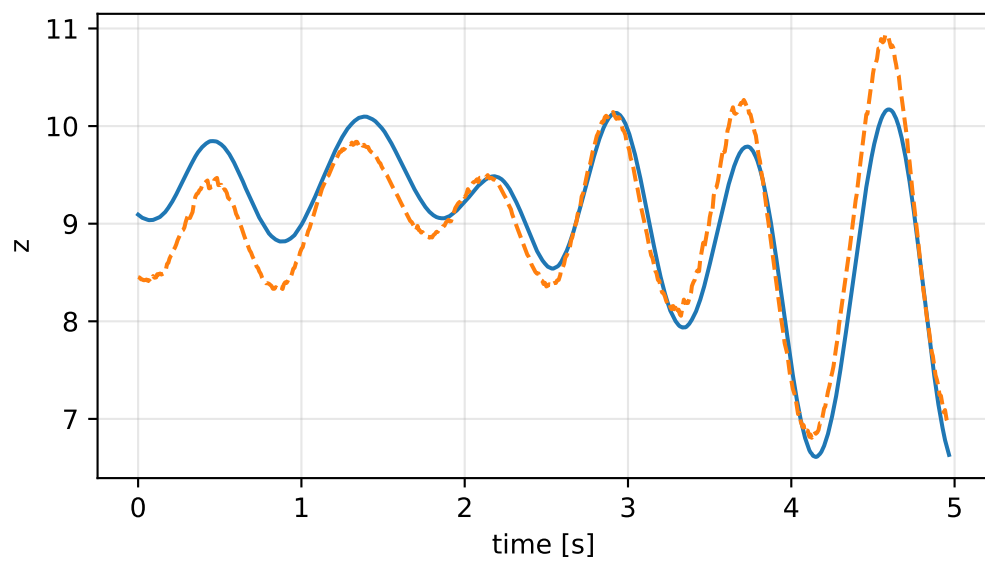
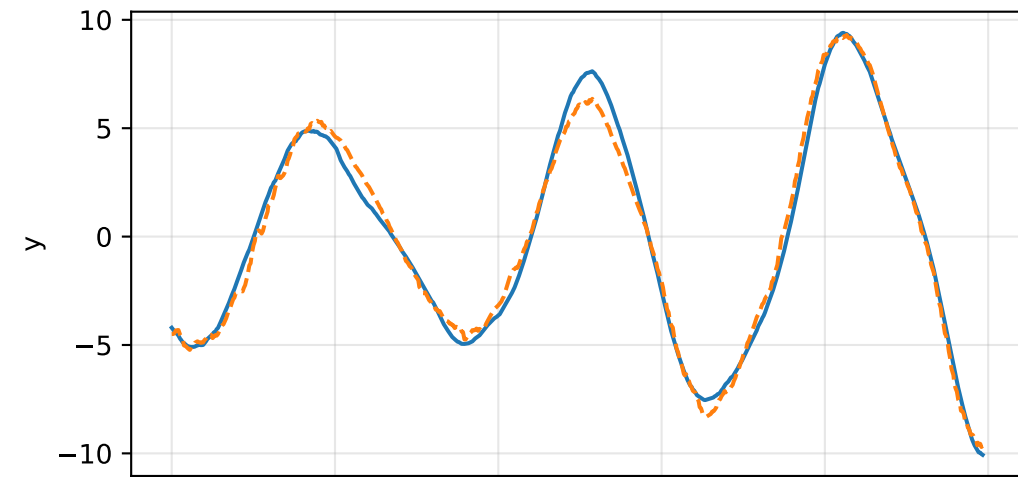
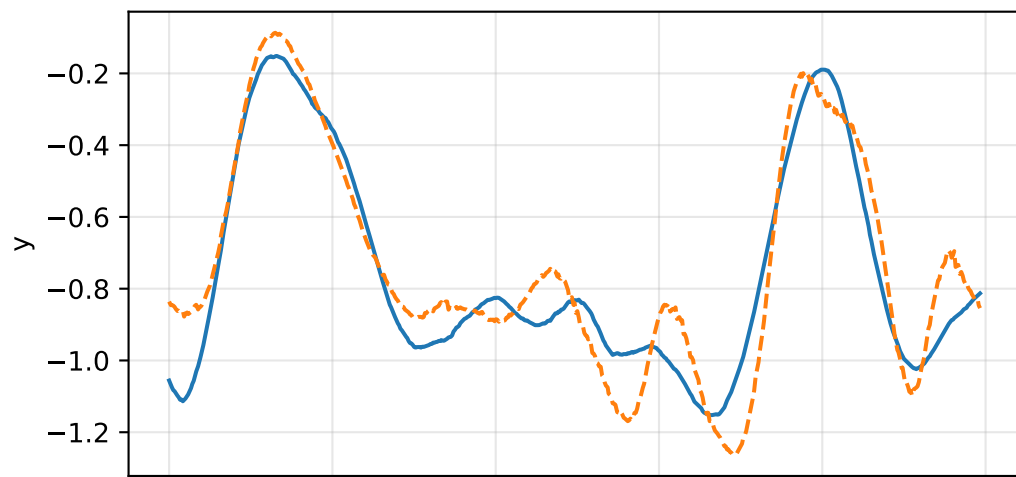
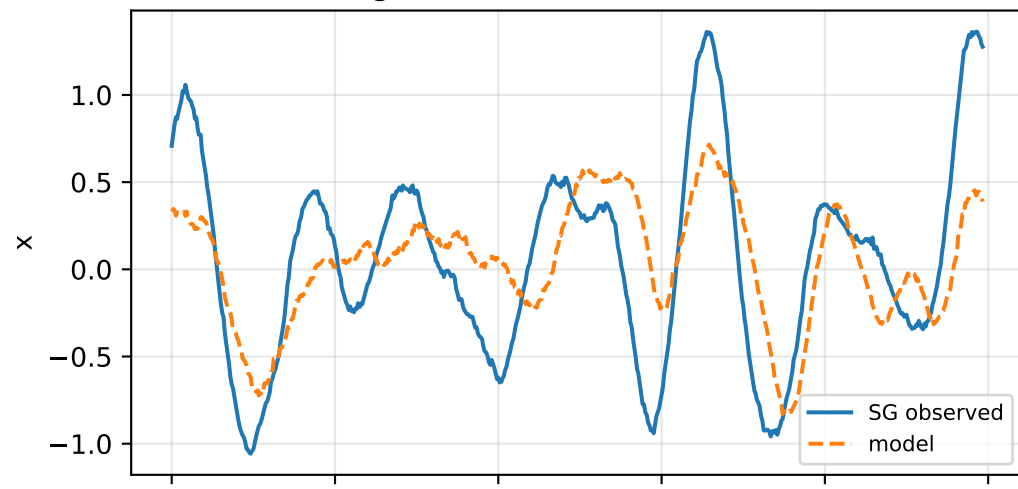


lag_pow2_depth_12: SG trajectory and derived motion

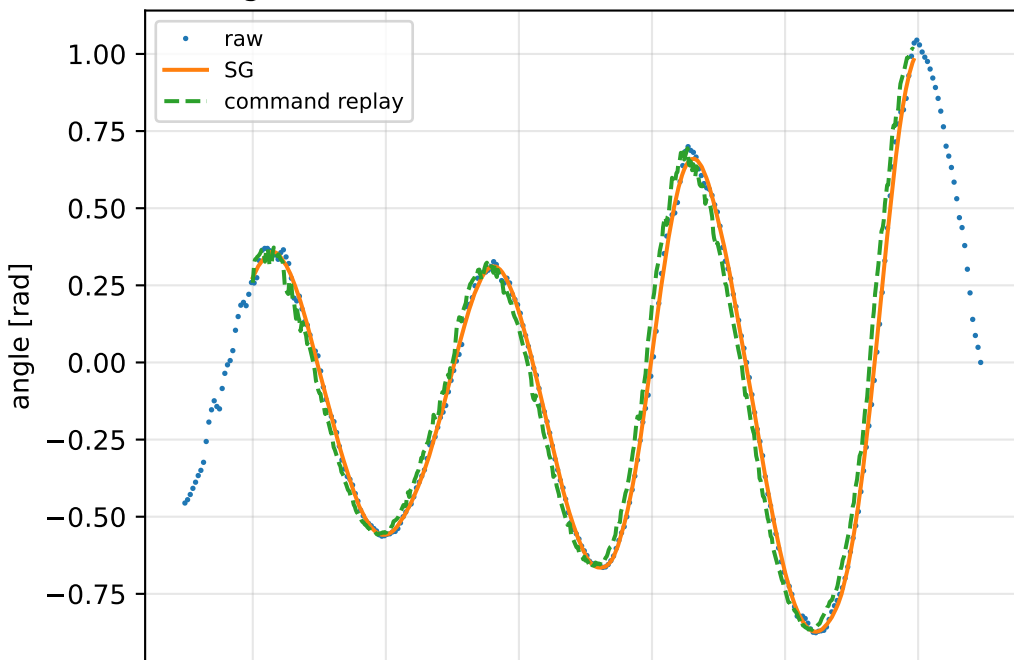


lag_pow2_depth_12: acceleration residual objective

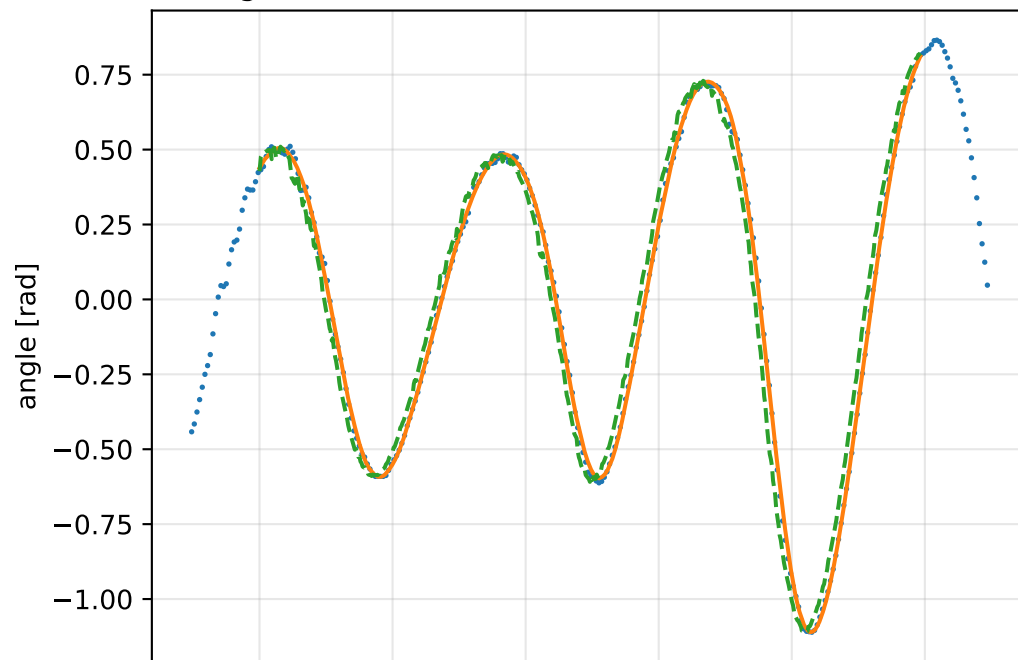
specific acceleration [m/s^2]angular acceleration [rad/s^2]

lag_pow2_depth_12: gimbal measurement audit (objective=measured_sg)

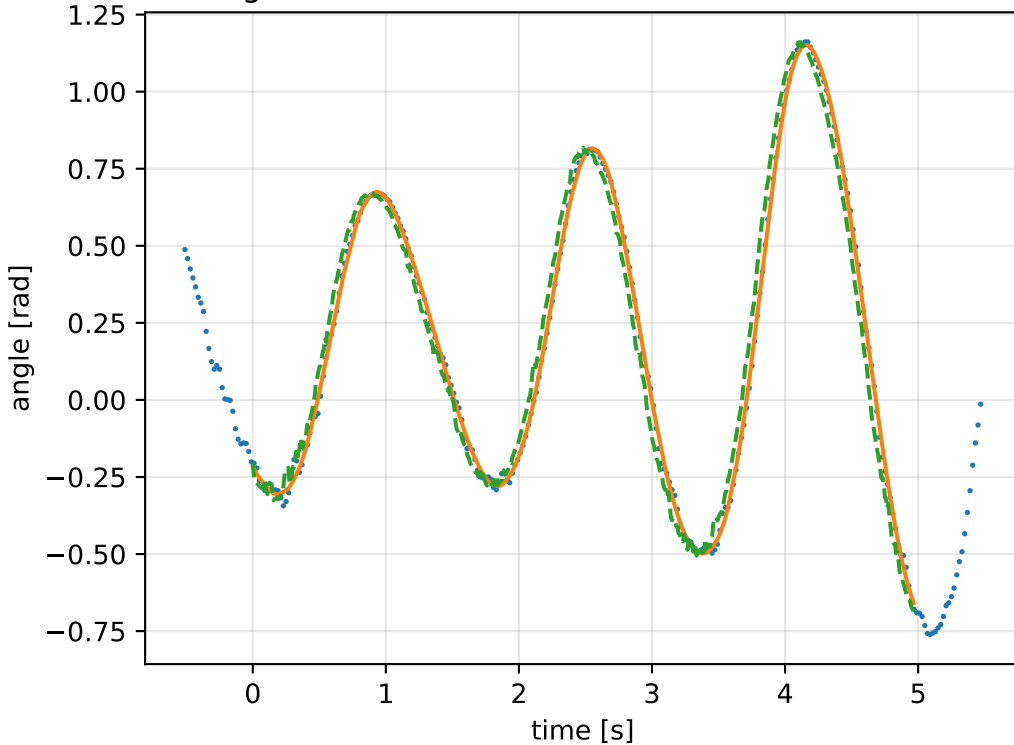
gimbal 1: RMSE=0.0759, max=0.2095 rad



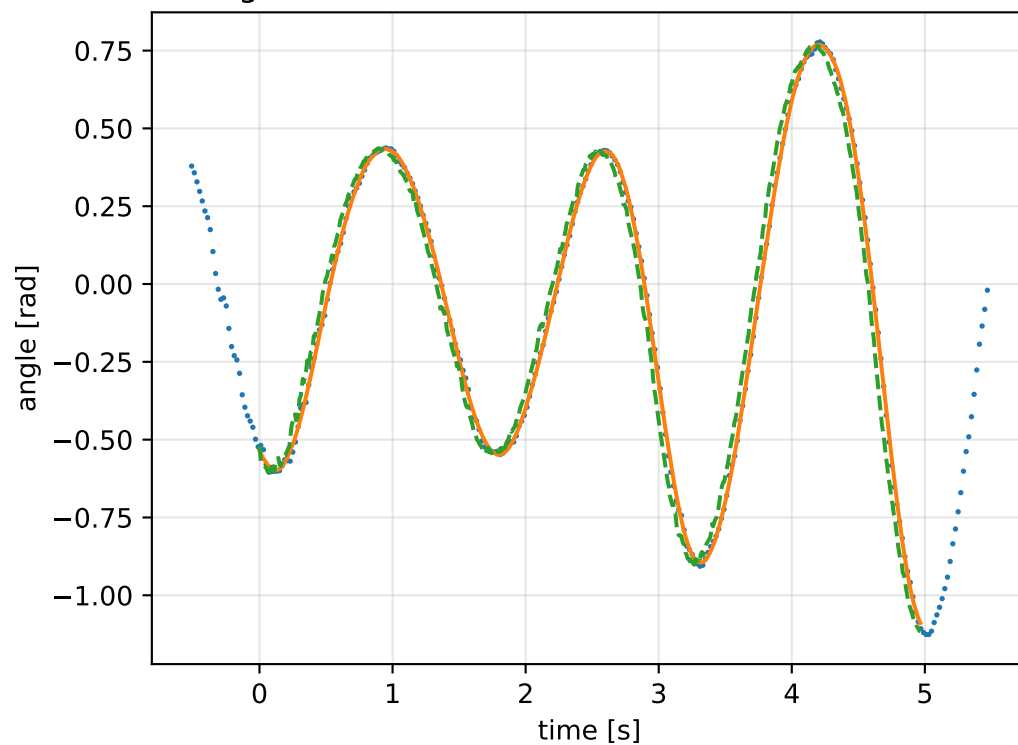
gimbal 2: RMSE=0.08099, max=0.2013 rad



gimbal 3: RMSE=0.07393, max=0.172 rad

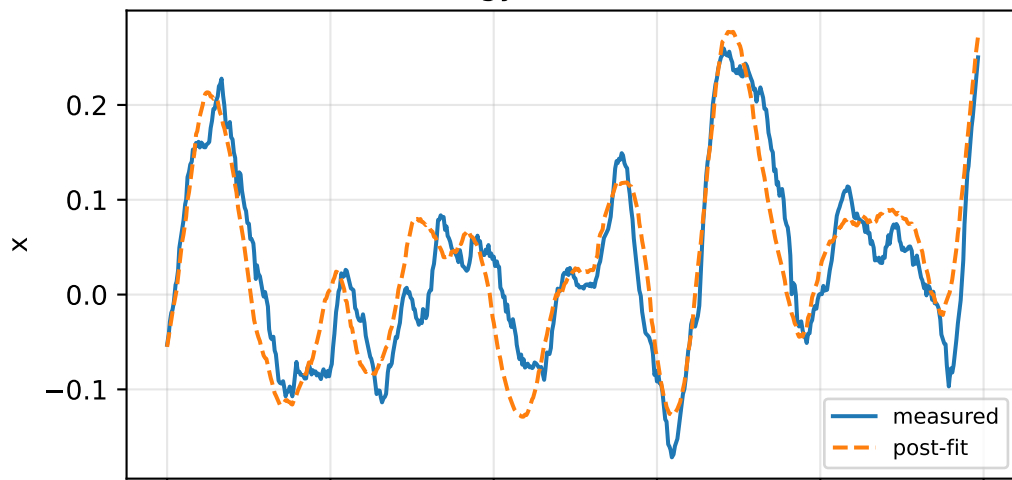


gimbal 4: RMSE=0.07131, max=0.1737 rad

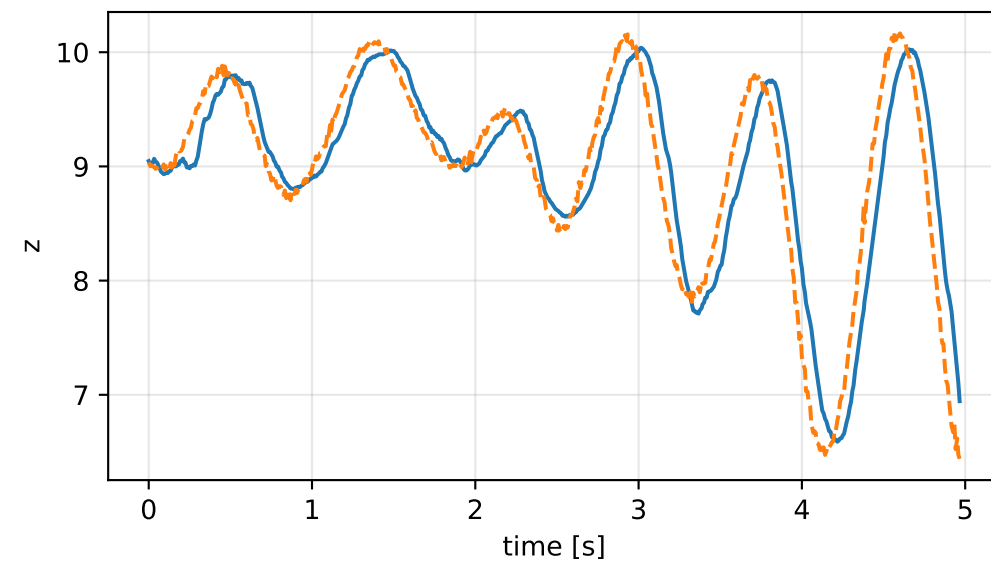
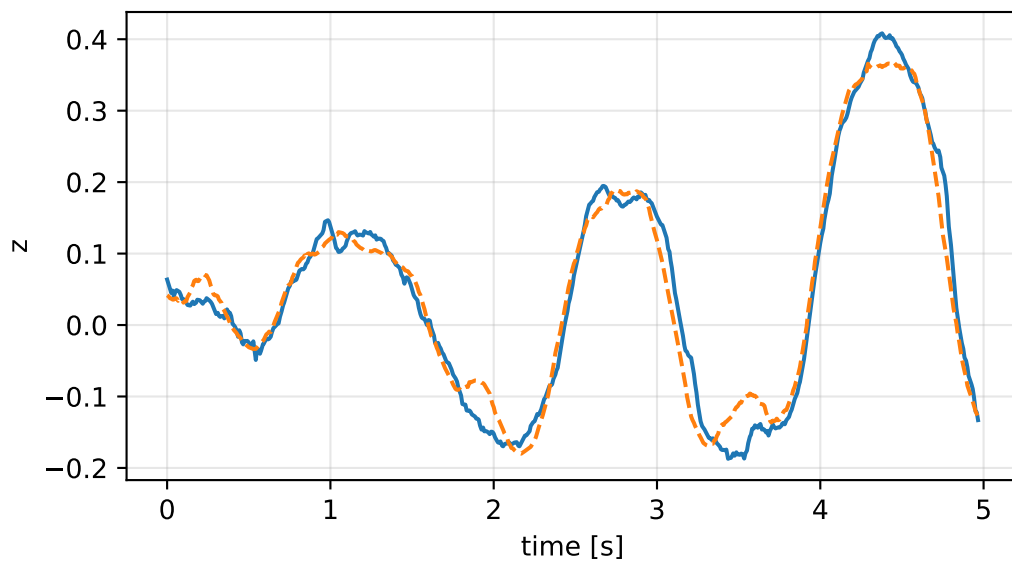
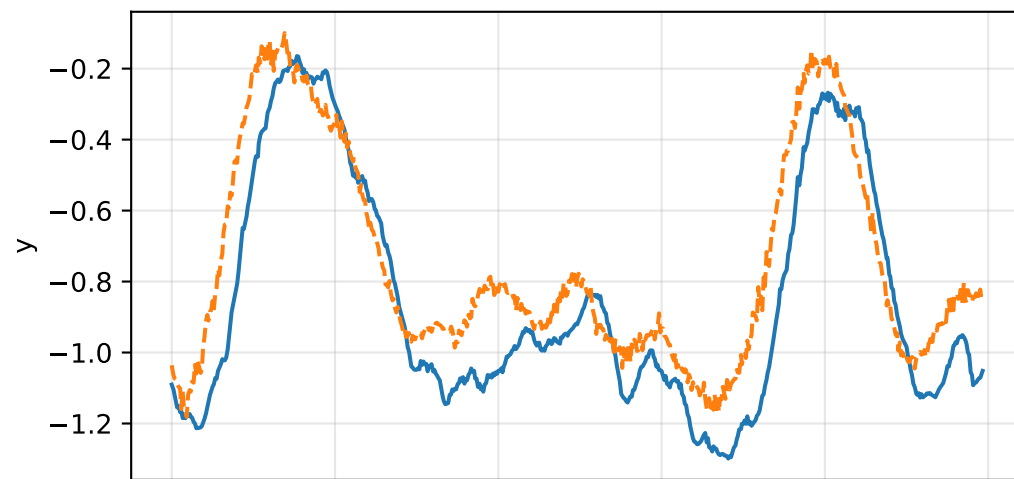
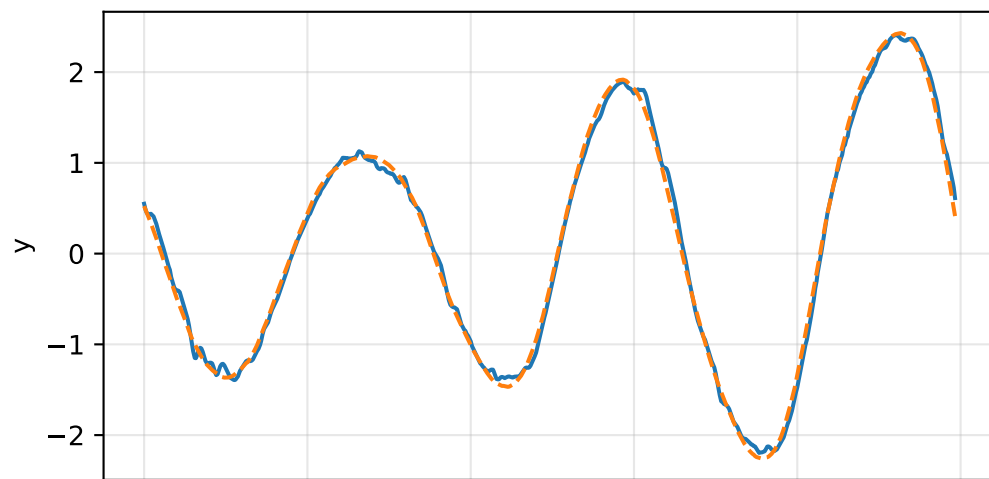
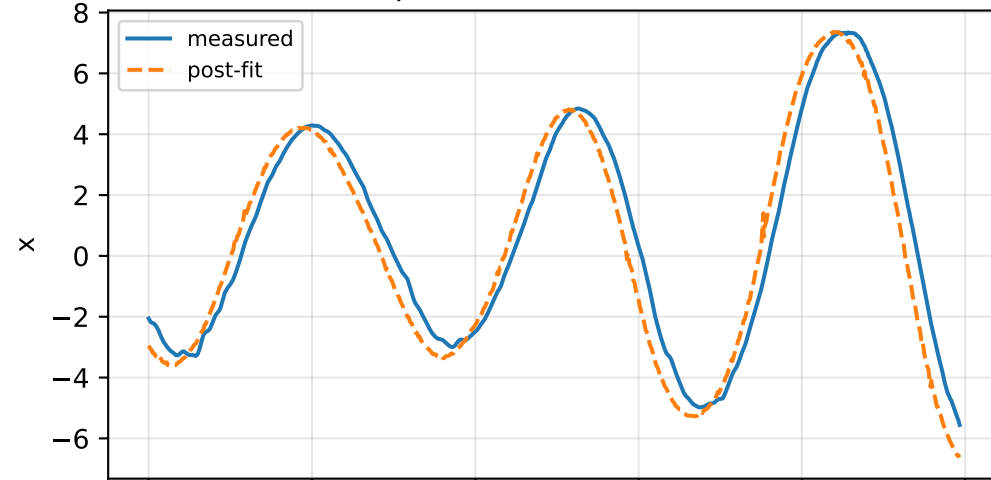


lag_pow2_depth_12: IMU comparison (diagnostic only)

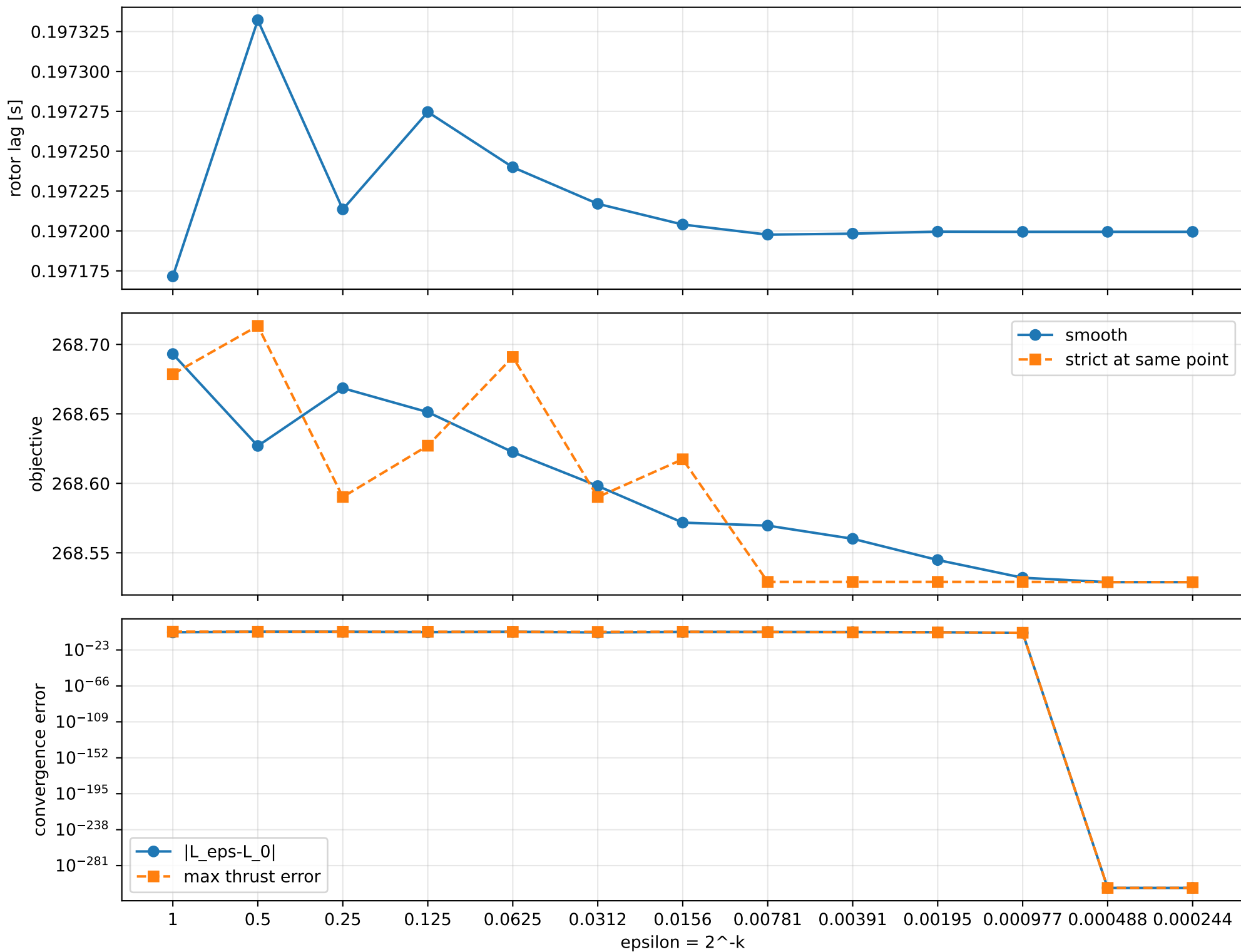
gyro [rad/s]



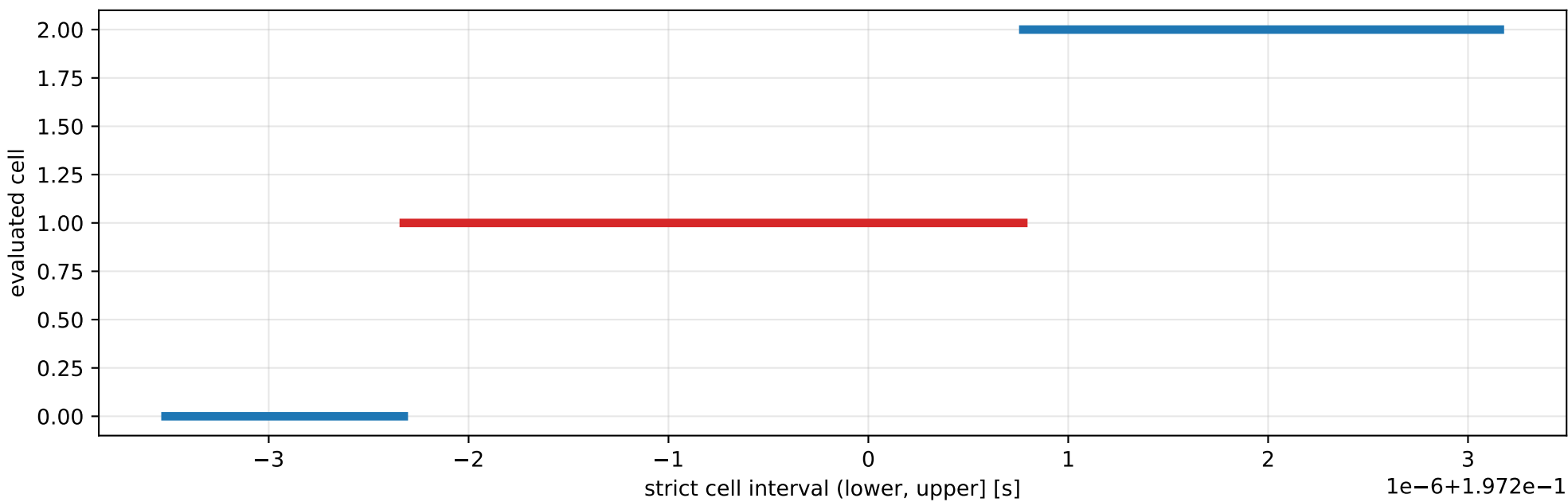
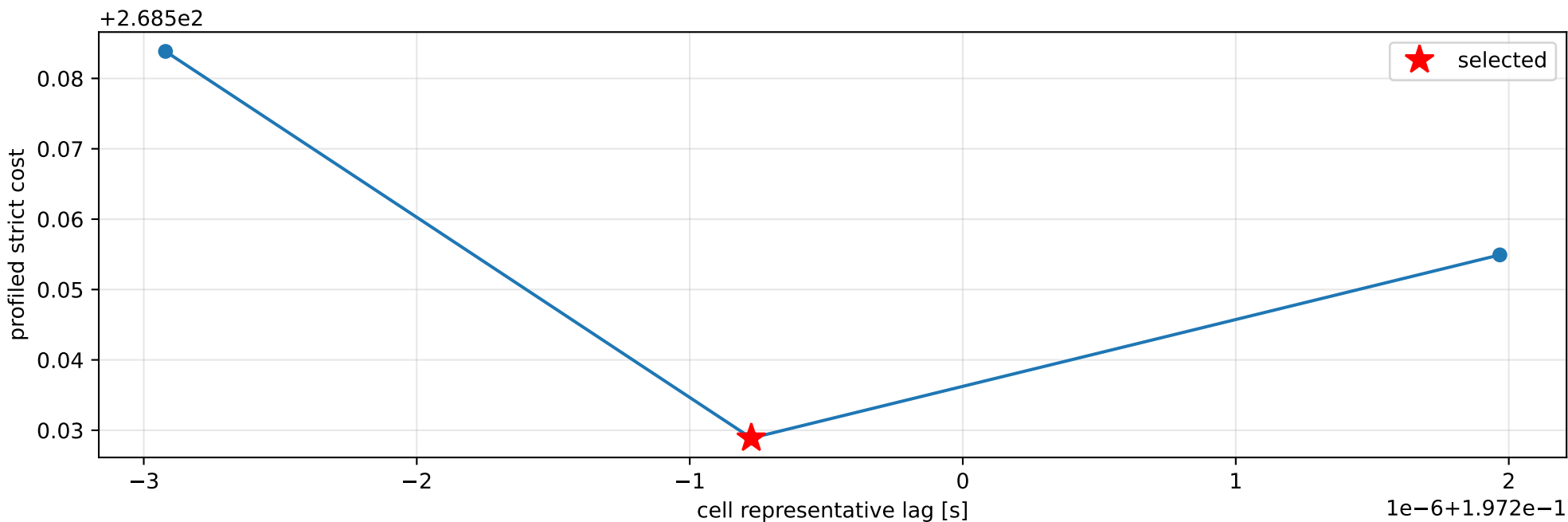
specific force [m/s²]



lag_pow2_depth_12: smooth-to-strict continuation



lag_pow2_depth_12: exact strict-ZOH lag cells



selected (0.197197676, 0.197200775] s; final smooth 0.197199468 s; data boundary=False

lag_pow2_depth_12: parameter estimate

Case: lag_pow2_depth_12

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 3.47573866

inertia [kg m²]:

[[3.53067572e-01 -4.27082465e-04 -5.73203906e-02]

[-4.27082465e-04 2.68867347e-01 3.31032087e-05]

[-5.73203906e-02 3.31032087e-05 5.96256142e-01]]

CoG body [m]: [0.01762796 -0.00411732 -0.04437564]

force effectiveness: [1.16405689 1.49895704 1.30895614 1.08526019]

scale-free J/m [m²]:

[[1.01580587e-01 -1.22875310e-04 -1.64915709e-02]

[-1.22875310e-04 7.73554554e-02 9.52407872e-06]

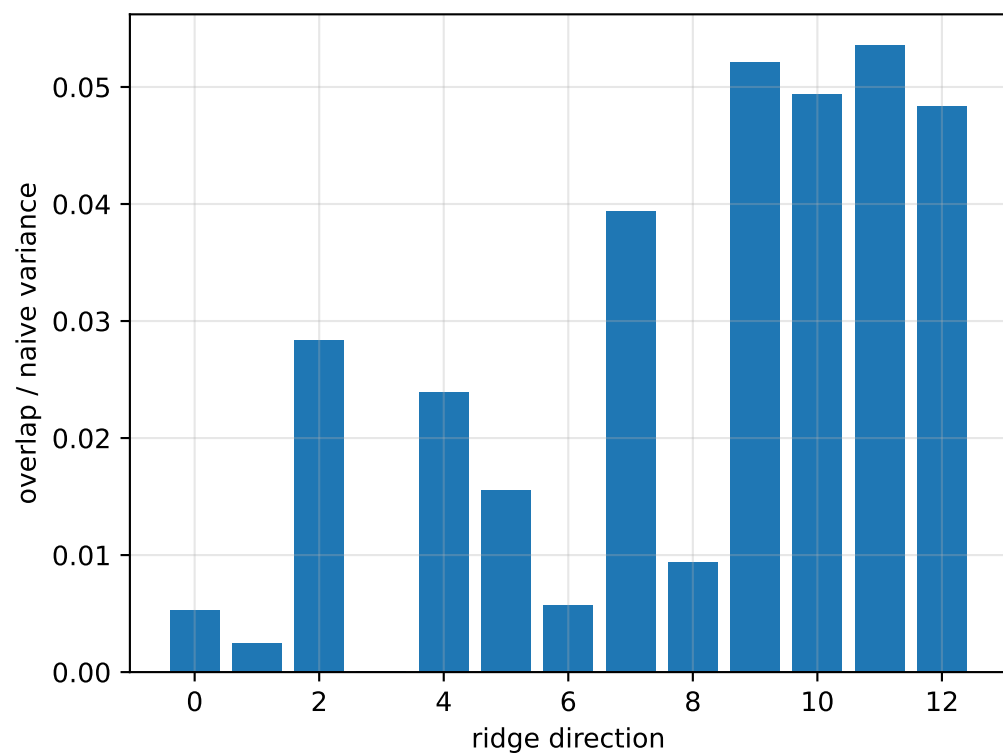
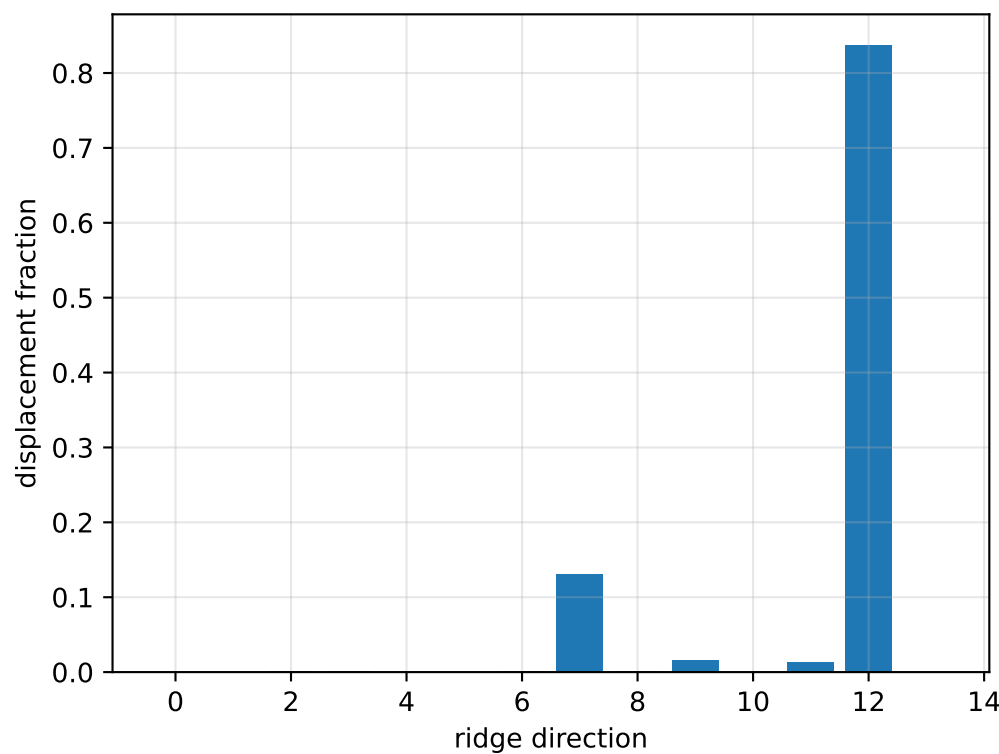
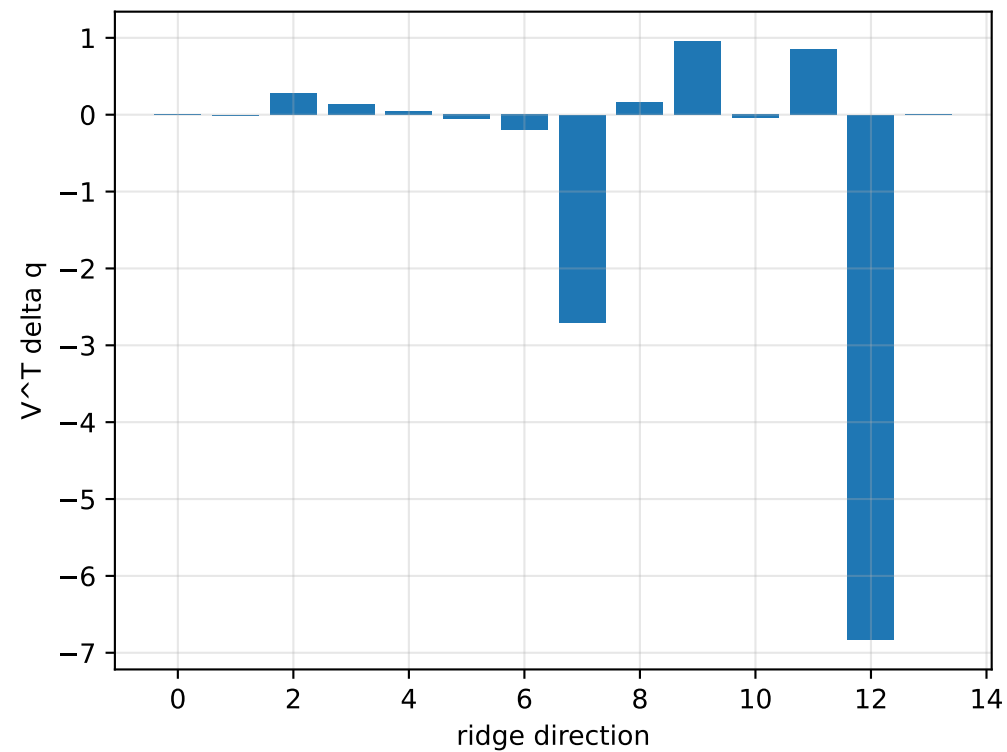
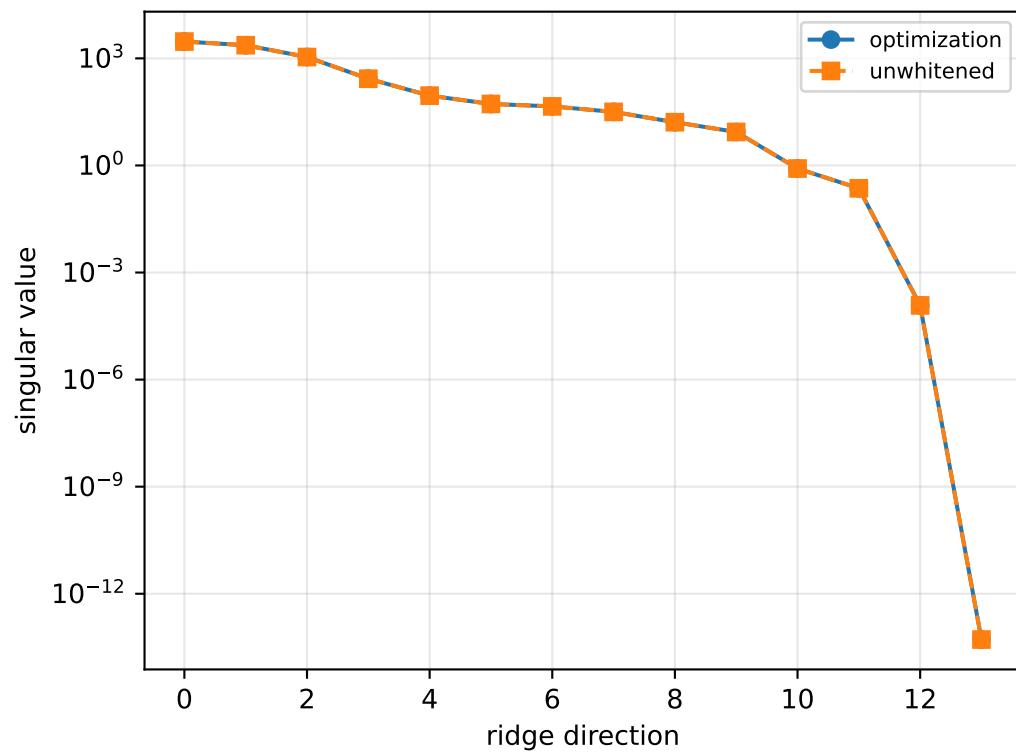
[-1.64915709e-02 9.52407872e-06 1.71548036e-01]]

scale-free f/m: [0.33490921 0.43126287 0.37659797 0.31223872]

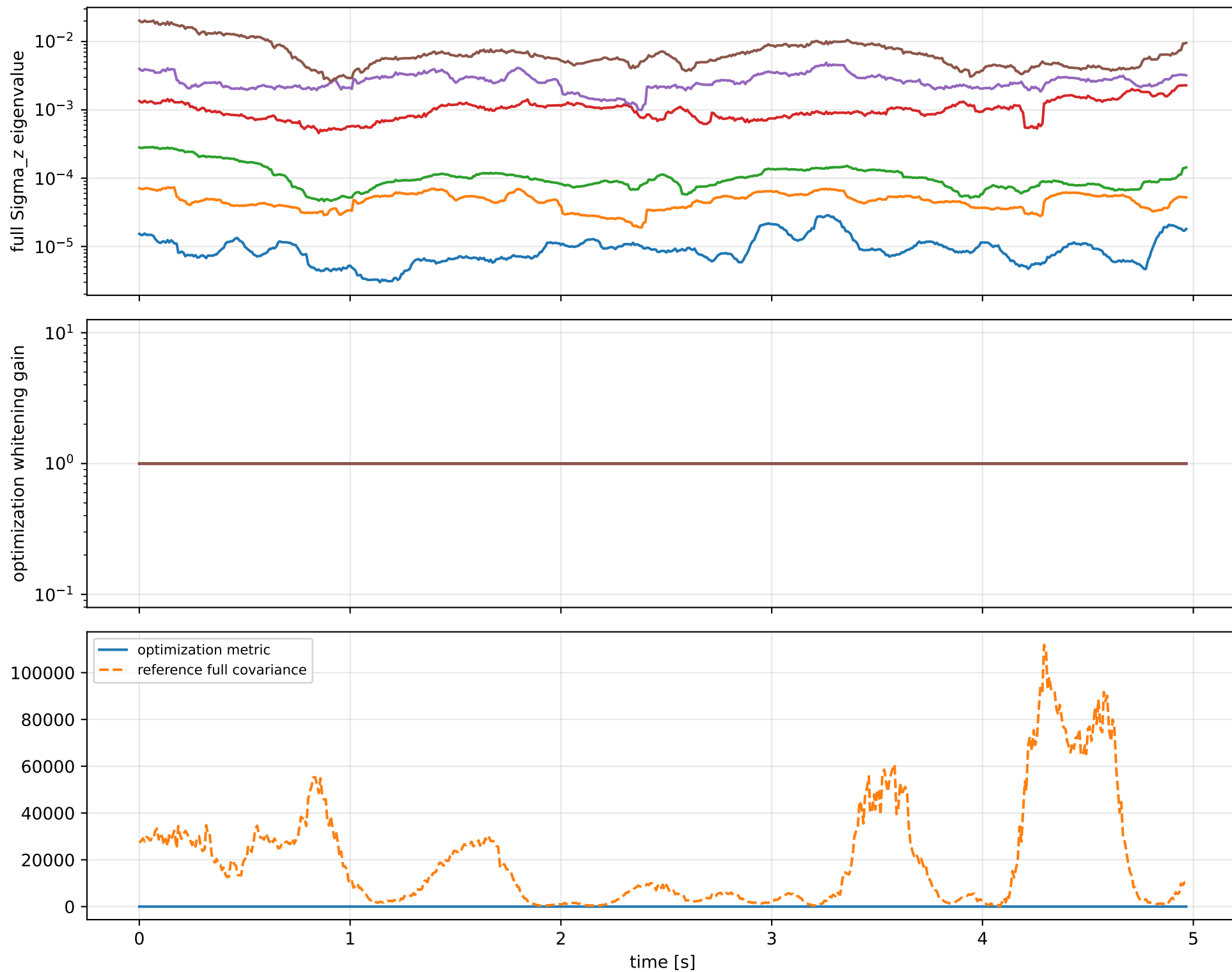
rotor lag cell: (0.197197676, 0.197200775] s

representative: 0.197199225 s

lag_pow2_depth_12: ridge spectrum (rank 13, nullity 1, $\|Jv\|=1.03e-13$)



lag_pow2_depth_12: optimization vs reference covariance



lag_pow2_depth_12: wrench / closure (force max $1.04\text{e-}14$, torque max $1.11\text{e-}15$)

